

Anastasiia Mokhonko

DATA SCIENCE & AI STUDENT

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“Aspiring Data Scientist and AI Specialist passionate about digital innovation, machine learning, and solving real-world problems.”

Summary

Machine learning engineer with hands-on experience across the full AI lifecycle, from research and model development to production deployment. Currently interning as an AI Product Engineer at Vanderlande (CDAO), building an automated code generation tool on Databricks and Azure targeting a 50–60% reduction in engineering time. Previously developed custom deep learning architectures for CT bone segmentation, achieving clinical-grade accuracy with 90% fewer parameters than the state-of-the-art baseline. Project portfolio spans NLP, computer vision, and MLOps across clients including Content Intelligence Agency, NPEC, NAC Breda, and ANWB. Additional background in bookkeeping and financial reporting for a large metal trading company. Applies GDPR, EU AI Act, and EU MDR compliance as standard practice.

Experience

Vanderlande – Central Data & Analytics Organization (CDAO)

Veghel, NL

AI PRODUCT ENGINEER INTERN

Feb 2026 – Jul 2026

- Building an end-to-end automated code generation tool eliminating 7–10 hours of manual work per Data Hub implementation, targeting 50–60% reduction in engineering time across Vanderlande’s enterprise data lake platform (Databricks, Azure).
- Implemented a production-grade Python pipeline parsing structured Excel/CSV mapping documents to auto-generate four standardized output files per Data Hub, validated across 11+ hubs spanning Finance, HR, Supply, and Technology domains.
- Integrated Meta Llama 3.3 70B via Databricks MLflow Deployments, combining LLM assistance with template-based generation; benchmarked against M365 Copilot achieving ~98% vs ~80% accuracy.
- Deployed an interactive Gradio UI via Databricks Apps for non-technical engineers; maintained a 59-test Python unit suite ensuring cross-domain reliability and regression coverage.
- Operating within a SAFe Agile Release Train, contributing to sprint planning, backlog refinement, and stakeholder demos.

Bone Segmentation for Surgical Planning – Confidential Medical Technology Company

Breda, NL

AI RESEARCHER, DEEP LEARNING ENGINEER

Sep 2025 – Jan 2026

- Developed custom ResU-Net deep learning architectures for automated radius and ulna bone segmentation from CT scans, achieving 86.9% test Dice score — within 3.1 pp of the nnU-Net state-of-the-art benchmark.
- Reduced model complexity by 90% (3.2M vs 31M parameters) and training time by 89% (7 hours vs 63 hours) while maintaining clinical-grade accuracy above the 85% acceptability threshold.
- Built and validated an 8-step preprocessing pipeline (voxel standardization, intensity normalization, patch extraction, HDF5 batching) across 162 CT cases from Radboud University Medical Center.
- Conducted 150+ training runs comparing U-Net, ResU-Net, V-Net, and nnU-Net architectures; delivered regulatory and ethical compliance framework for EU MDR Class IIa medical device deployment.

Emotion Detection NLP Pipeline – Content Intelligence Agency

Breda, NL

NLP ENGINEER, MLOPS ENGINEER, DEVELOPER

Feb 2025 – Jun 2025

- Developed NLP pipeline for emotion analysis in French reality TV; implemented and compared Naive Bayes, Logistic Regression, LSTM, BERT, and CamemBERT models; applied XAI techniques (LIME, SHAP) for interpretability.
- Deployed production-ready system classifying 6 emotions in French video/audio content with 44.7 requests/second capacity.
- Built Azure ML pipelines with automated training, retraining, and deployment using Docker and CI/CD; developed FastAPI endpoints and Streamlit interface with blue-green deployment for zero-downtime updates.

Root Segmentation & Robotic Control – NPEC Netherlands Plant Eco-phenotyping Centre

Breda, NL

DEVELOPER, COMPUTER VISION ENGINEER

Nov 2024 – Dec 2024

- Built computer vision pipeline using U-Net for plant root segmentation and graph theory (NetworkX, Dijkstra) for measurement.
- Developed PID and Reinforcement Learning controllers (Stable Baselines 3) for Opentrons OT-2 robotic microbial application.
- Achieved 99% time reduction in root analysis workflow; delivered high-accuracy solution to research client.

Cybersecurity Awareness in Small Hotels – Policy Research for DigiWerkplaats

Breda, NL

ANALYTICS TRANSLATOR, TEAM LEAD, RESEARCHER

Sep 2024 – Oct 2024

- Led research team of 5 investigating employee cybersecurity awareness impact on hotel security using mixed-methods approach.
- Designed and analyzed surveys (Qualtrics) and interviews; ensured GDPR compliance and research ethics approval.
- Found moderate positive correlation ($r=0.456$) between training frequency and security adherence; recommended quarterly training.

ANWB Safe Driving Application – ANWB Dutch Automobile Association

DEVELOPER, DATA ANALYST

Breda, NL
May 2024 – Jun 2024

- Developed Streamlit application analyzing driver behavior and road safety using ANWB datasets and time series analysis.
- Implemented incident classification models with EU AI Act compliance and Agile project management methodology.

Farmer’s Buddy: Potato Blight Detection AI

DEVELOPER, DATA SCIENTIST

Breda, NL
Feb 2024 – Apr 2024

- Built AI-driven mobile application for potato blight detection using MobileNetV2 transfer learning, achieving 95.3% accuracy.
- Applied Grad-CAM for explainable AI and conducted think-aloud usability studies to optimize user experience.

Goal Prediction Analytics – NAC Breda Football Club

DATA ANALYST, MACHINE LEARNING ENGINEER

Breda, NL
Nov 2023 – Jan 2024

- Developed machine learning model analyzing 16,535 players to predict goal-scoring potential with 95%+ accuracy.
- Delivered GDPR-compliant recruitment recommendations using linear regression and correlation analysis.

Human Trafficking Data Analysis Dashboard – UN SDG 16

DEVELOPER, DATA ANALYST

Breda, NL
Sep 2023 – Nov 2023

- Designed interactive Power BI dashboard analyzing global trafficking patterns by age, gender, exploitation type, and region.
- Calculated temporal correlation (r=0.68) and delivered policy recommendations for prevention strategies supporting SDG 16.

Confidential – Large Metal Trading Company

PART-TIME FINANCE ASSISTANT

Netherlands
Nov 2023 – Present

- Supporting bookkeeping and financial reporting for a large-scale metal trading operation.
- Maintaining financial records, preparing reports, and developing practical accounting knowledge under the guidance of a finance professional.

Education

Breda University of Applied Sciences

BACHELOR OF APPLIED DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Breda, Netherlands
Sep 2023 – Present

ITSTEP Academy

PYTHON DEVELOPER COURSE

Kyiv, Ukraine
2022 – 2023

- Completed 6-month intensive course covering Python, OOP, algorithms, SQL/NoSQL, Django, Git, and cloud technologies.

Extracurricular Activity

Degree Programme Committee (DPC) – Data Science & AI Programme

STUDENT REPRESENTATIVE

Breda University of Applied
Sciences, Breda, NL
2024 – 2026

- Selected as one of three student representatives to collaborate with faculty on curriculum quality, course delivery, and student experience.
- Act as liaison between student body and programme management, representing student perspectives in policy and assessment decisions.

Nakormim Bali – Orphanage Support Organization

VOLUNTEER & MONTHLY DONOR

Bali, Indonesia
2022 – Present

- Support post-COVID-19 relief for orphanages through monthly donations, educational sessions, and coordinated activities for children.

Technical Skills

Programming	Python, SQL, HTML/CSS, LaTeX
ML/AI Frameworks	PyTorch, TensorFlow, Scikit-learn, Transformers (HuggingFace), BERT, CamemBERT, U-Net, LangChain, RAG
MLOps & Cloud	Azure ML, Azure DevOps, Databricks (Unity Catalog), MLflow, Docker, CI/CD, Git, FastAPI, Gradio, SAAge Agile
Data & Visualization	Power BI, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Apache PySpark, SQL (PostgreSQL, MySQL)
Specializations	NLP, Computer Vision, Deep Learning, LLM Integration, Time Series Analysis, XAI (LIME, SHAP), Prompt Engineering
Tools & Platforms	VSCode, Jupyter, Git/GitHub, Jira, Confluence, Figma, Trello, Notion

Languages

Native	Russian, Ukrainian
Professional	English (C1 - Advanced)
Intermediate	French (B2 - Upper Intermediate)
Beginner	Indonesian (A2 - Elementary)